

Strategy Brief #1

Climate Risk & Coastal Infrastructure in the Gulf of Guinea.



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Introduction

Rapid urbanization in coastal cities of the Gulf of Guinea, combined with **land subsidence** and **sea-level rise**, is significantly increasing the **exposure of populations and infrastructure to flooding and long-term environmental risks**.

These dynamics are particularly critical in countries such as **Cameroon, Nigeria, Gabon, and Angola**, where urban growth outpaces infrastructure planning.



Key Challenge

By 2050, the interaction between climate change and human-induced subsidence could substantially amplify risks to:

- Urban infrastructure (roads, ports, housing)
- Water and sanitation systems
- Economic activities in coastal zones

Despite this, limited data availability and weak planning frameworks hinder effective decision-making and investment prioritization.

Strategic Insights

- **Human activities (urbanization, groundwater extraction)** are a major driver of land subsidence, often exceeding natural processes.
- **Risk exposure is accelerating**, not only due to climate factors but also due to **unplanned urban expansion**.
- Without intervention, **economic losses and infrastructure damage could increase exponentially** by 2050.
- Current planning approaches are often **reactive rather than anticipatory**.

Strategic Implications

- Coastal risk is not only an environmental issue but a **strategic economic and development challenge**.
- Governments and investors face **increasing financial risk** due to infrastructure exposure.
- There is a need to **shift from short-term responses to long-term resilience planning**.

Recommendations

- **Integrate climate and subsidence risk into urban planning and infrastructure investment decisions**
- **Strengthen data systems** using geospatial technologies for risk monitoring
- **Prioritize resilient infrastructure development** in high-risk zones
- **Promote cross-sector coordination** (water, urban planning, environment)
- **Support policy frameworks** that encourage sustainable land and water management

Value Proposition

A strategic, data-driven approach to coastal risk management can:

- Reduce long-term infrastructure costs.
- Improve investment efficiency.
- Enhance resilience of rapidly growing urban populations.